The Unreasonable Effectiveness of Simplicity in AI

(Why less is more, even in the age of LLMs)

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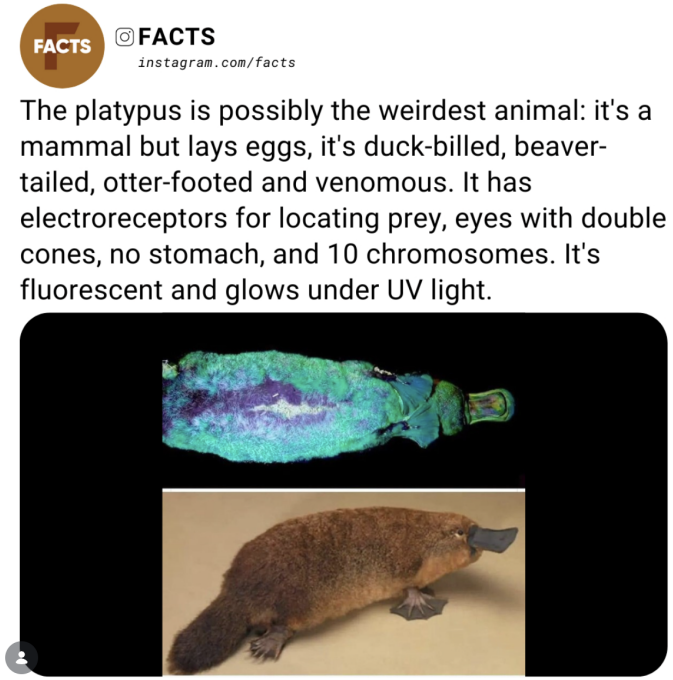
February 2026

Australia: Land of Weird Ideas



- **A Larrikin Culture:** ideas found elsewhere are tested/broken here.
  - Here, even water is a contrarian (swirls clockwise).
- Oz's vastness demands rugged, simple, and efficient tech.
- **Embrace the Ozzieness:** Explore off the beaten track.
- A great place to explore **heresies**.

The Prevailing Myth

- **The Narrative:** Conventional AI "failed" and was "repaired" by LLMs.
- **The Reality:** Pre-LLM AI was finding elegant shortcuts.
- We were discovering deep insights into human cognitive styles.
- These insights are being swept away in the urge to commercialize.
- Is "Bigger" always "Better," or just "More Expensive"?

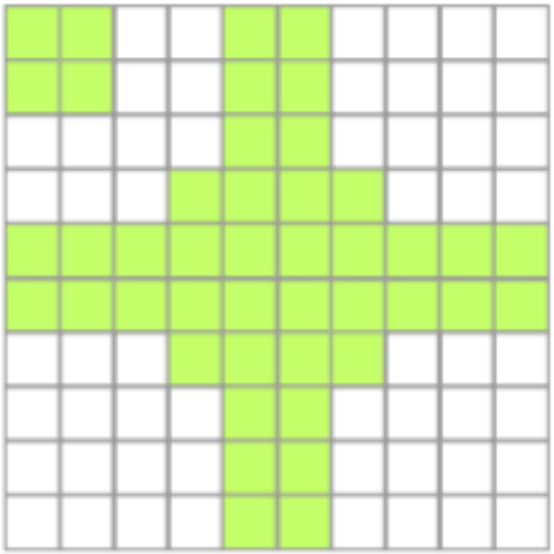
A Millennial Tradition of Simplicity

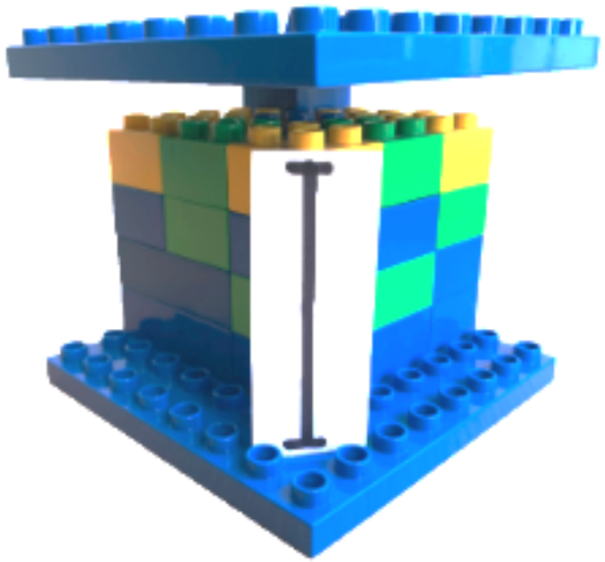
- **130 AD (Ptolemy):** Explain phenomena by the simplest hypothesis.
- **1300s (Occam):** Entities must not be multiplied beyond necessity.
- **1984 (JL Lemma):** Random projections preserve pairwise distances [8].
- **1997 (Feature Selection):** We can often ignore 80% of features [9].
- **2002 (Backdoors):** Setting a few variables cuts exponential time [10].
- **2025 (Compact AI):** Performance $\approx$ LLMs at lower cost [2].

Benchmark	Total Vars (N)	Backdoor Size ($\|A'\|$)
logistics.d	6,750	12
3bit-adder_32	8,704	53
pipe_07	1,685	23
qcp_15	3,375	290
map_v6	3,929	167

Why Simplification Studies are Rare

- In a survey of 229 SE papers on LLMs, only **5%** used baselines [11].
- Biased against simplicity [12]
- Without a "Simpler" baseline, we aren't doing Science.
- We must support alternatives for strategic, auditable tasks.





Mapping AI to Human Cognition: RDR

- Paul Compton, **Ripple Down Rules (RDR):** Paradigm shift in Knowledge Acquisition.
- Avoids the "KA Bottleneck" by supporting **Repair in Context**.
- Experts justify why *this* case is different.
- Maintenance and acquisition become the same task [1].

Table 1
Summary data for 7 sample KBS

	Total cases interpreted	Total rules added	Months in use	Cases per month
A	1,490,767	1061	29	51,406
B	1,333,598	1091	18	74,089
C	1,205,566	339	28	43,056
D	419,555	123	24	17,481
E	271,371	1036	21	12,922
E′	187,848	9307	29	6,478
F	46,176	2021	23	2,008

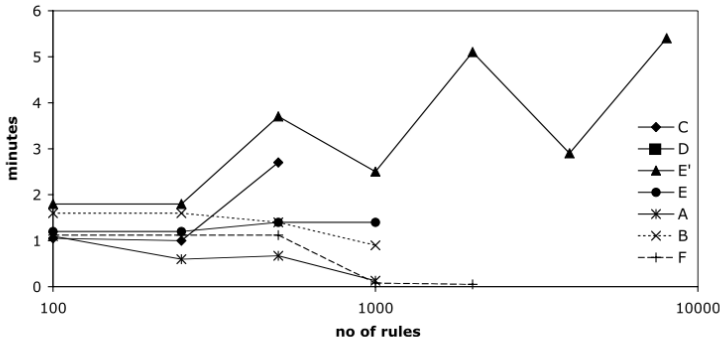


Fig. 5. The average time taken to add a rule at different KB sizes for some sample KBS.

Better than the Masters: PIGS

- Timm: **AUSPIG Expert System:** Back-end for pig growth simulation.
- **The Result:** It outperformed the humans who wrote its rules.
- Achieved ≈ 10% improvement in profitability ($\$/m^2/day$).
- **Key point:** Humans provided rules; the system provided consistency.
- Fielded globally (USA, Holland, Australia) [7].

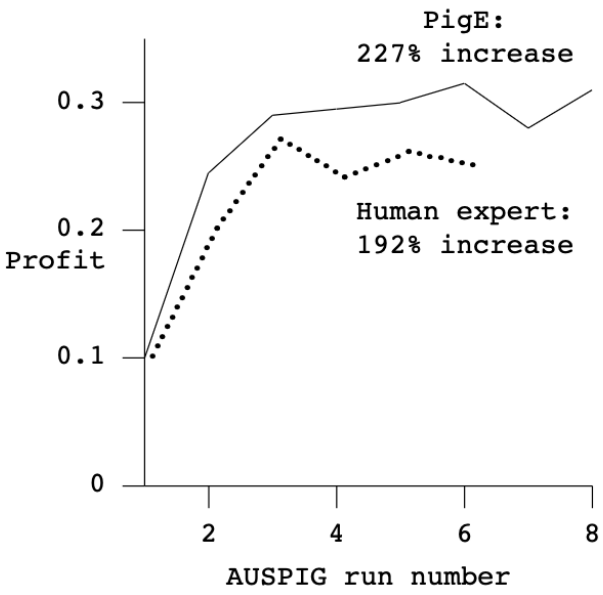


Figure 2. Comparison of human expertise vs PigE

Case Study: NB.awk

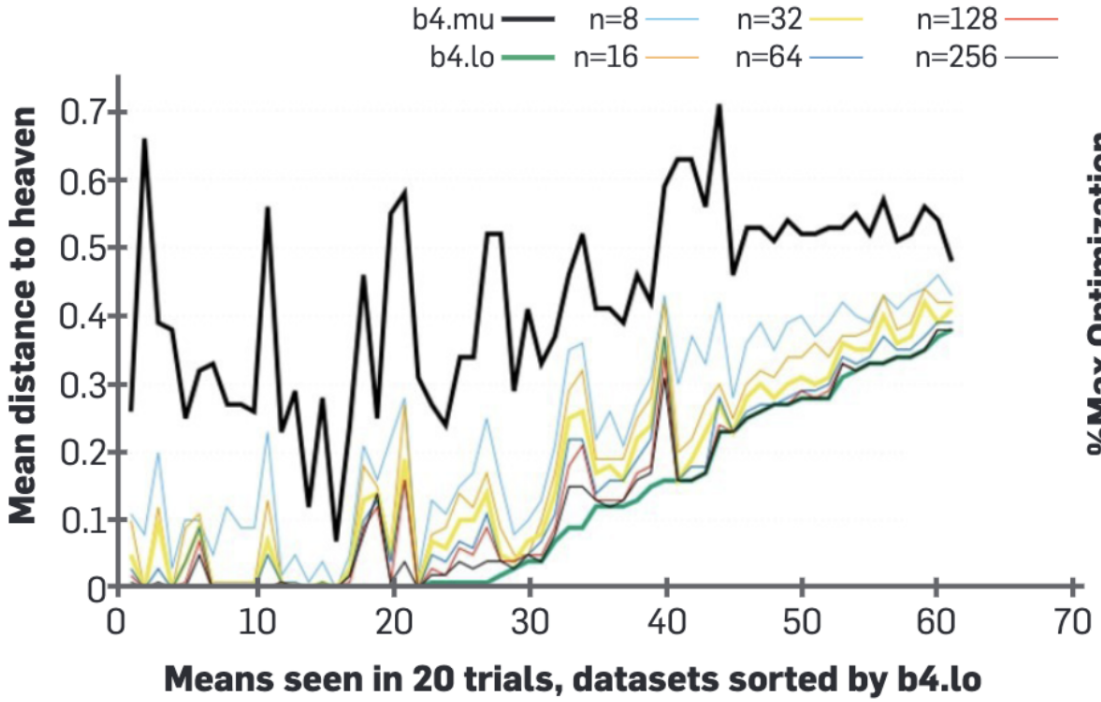
- That moment when you realize just how easy AI can be...

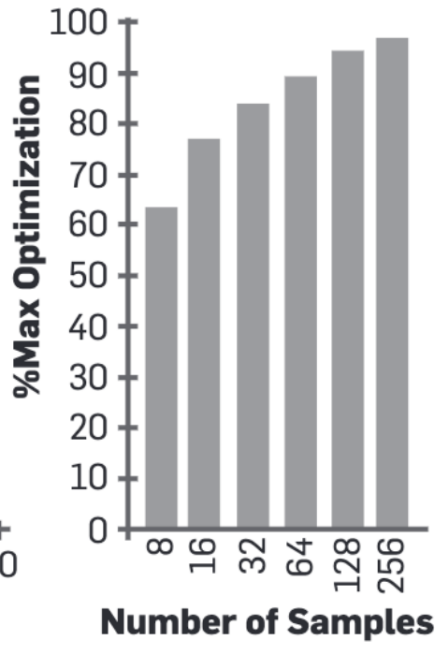
```
function train(i,c) {
  Total++; c=$NF; Classes[c]++
  for(i=1; i<NF; i++) {
    if($i=="?") continue
    Freq[c,i,$i]++; if(++Seen[i,$i]==1) Attr[i]++ }}

function classify(i,c,t,best,bestc) {
  best=-1e30; for(c in Classes) {
    t=log(Classes[c]/Total)
    for(i=1; i<NF; i++) {
      if($i=="?") continue
      t+=log((Freq[c,i,$i]+1)/(Classes[c]+Attr[i])) }
    if(t>best) { best=t; bestc=c }} return bestc }
```

The Case for Compact AI

- Strategic tasks (auditable) require non-LLM tools [2, 4].
- Timm: **EZR**: **E**a**z**y Reasone**r**
- Stochastic learner exploiting software's sparsity.
- Achieves 90% of peak results with only **32 samples**.





LLM vs. Compact AI: A Comparison

Feature	Large Language Models	Compact AI (EZR/RDR)
Resource Req.	Massive (H100s/GWs)	Minimal (Laptop/AWK)
Reasoning	Probabilistic/Opaque	Causal/Auditable
Data Needed	Terabytes	Dozens of labels [3]
Maintenance	Retraining ($$$)	Local Repair (RDR)

Conclusion: Simplicity is a Superpower

- Simplicity isn't just "easier"; it's robust and auditable.
- Don't prove absence of errors; **herd** toward goals [4].
- Let's stop building "Bigger" and start building "Smarter."
- **Next Step:** Compare your next LLM project against 200 lines of LUA.

References

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- **[9]** Kohavi, R. & John, G. "Wrappers for feature subset selection," *AIJ*, 1997.
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- **[11]** Yang, Y. et al. "Survey on deep learning for SE," *ACM CSUR*, 54(10), 2022.
- **[12]** Adams, G.S. etl. "People systematically overlook subtractive changes," Nature, 592(785), 2021.